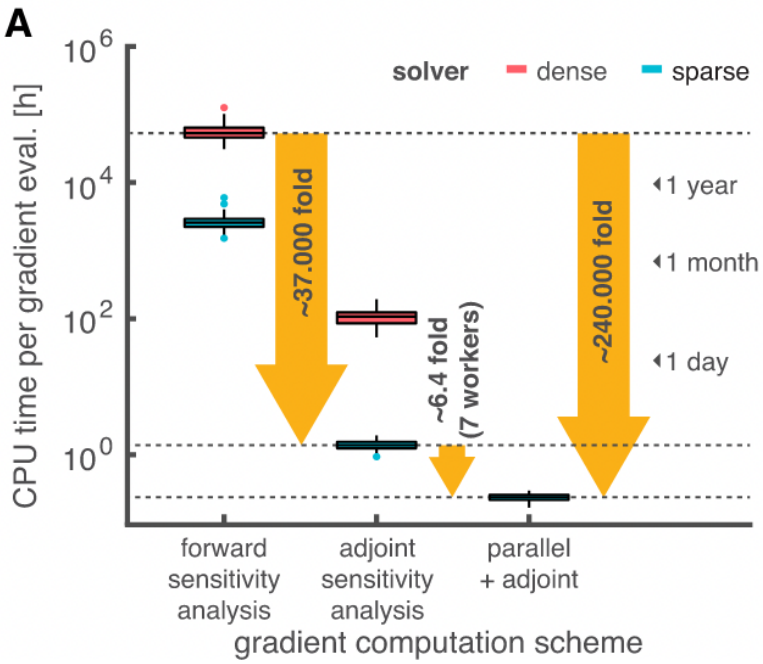
Toward large scale parameter estimation



- >4,100 unknown parameters
- Model trained using viability data from 120 cell lines x 7 different drugs x 9 concentrations
- 6,900 conditions used (5-fold cross-validation)
- Problem: gradient evaluation of objective function would take >6 CPU years!
- Combination of approaches led to > 5 order of magnitude reduction in computation time
  - Sparse linear solver for ODE integration
  - Adjoint sensitivity analysis
  - Coarse-grained parallelization